INFO 248 Project 2 Guidelines and Requirements

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Overview

This project is an opportunity for you and your partner to explore and analyze an aspect of a topic or domain using data science techniques from class. ***The choice of topic/domain and research question as well as the modeling techniques you use is up to you***. Then, you will need to find a dataset that will support your analysis. ***Note: you may not use a dataset that was presented in any class material.***
For this project, you and your partner will work collaboratively to produce a high quality report based on your analysis of a data set using multiple linear regression. A collaboration survey is provided as a means to communicate to the instructor any issues you are having with your partner. Note this is a graded component. A lack of collaboration may result in down-grading of an individual's grade on the project. Please contact the instructor if you have any questions about any part of the project including collaboration. See the Collaboration section below.

You will select ***at least two modeling techniques*** from class to apply to your data to investigate your main question(s). ***The same formatting requirements*** from project 1 apply to project 2.

As with project 1, you will submit an Rmd file as well as a pdf report that was produced from knitting your Rmd file. See Files to submit below.

There are t**wo milestones** where you and your partner will report on the progress you are making. The final project will be due at the end of the semester. A collaboration survey is

provided as a means to communicate to the instructor any issues you are having with your partner. Note this is a graded component. A lack of collaboration may result in down-grading of an individual's grade on the project. Please contact the instructor if you have any questions about any part of the project including collaboration.

Project 2 Timeline

Mar 14	Project 2 assigned	-
Apr 6	*Project 2 milestone 1 due	10 pts
Apr 19-25	*Partner eval 2 (due Apr 25)	5 pts
Apr 27	*Project 2 milestone 2 due	10 pts
May 12	*Project 2 due	60 pts
	Total	85 pts

* graded component

Project Milestones

In order to facilitate your project development, there will be two project milestones where you will submit materials. Note that there are ***no time extensions*** for these milestone assignments; you submit what you have.

Milestone 1:
The requirements and grading information can be found in the doc: INFO 248 Project 2 Milestone 1 Requirements.pdf

Milestone 2:
The requirements and grading information can be found in the doc: INFO 248 Project 2 Milestone 2 Requirements.pdf

Project 2 Required Sections (Final Submission)

These sections are required:

- **Overview**
    - Give a brief overview of the domain and aspects of that domain that you will study in your analysis. This is to provide context for the reader.

    - State your main research question.
- **Data Description and Processing**
    - Explain where the data set comes from.
    - For each variable,include:
    - The name of the variable.
    - A brief description of what it measures.
    - The data type: numeric or categorical. If categorical, include the names of the levels.
    - Identify the dependent variable.
    - State the size of the data before and after any cleaning, such as removing missing values and/or outliers. Include the count of observations that were removed.
    - Mention if you removed any outliers. State how many were removed.
- **Data Exploration**
    - You must provide a plot and table to describe the distribution of the dependent variable.
        - Required: a histogram with density curve and a table with descriptive stats:min, max, median, mean, SD (quartiles or quantiles optional).
    - For each predictor that you consider to be important or interesting, provide a plot and table that describes its distribution. This predictor or predictors will typically be mentioned in your hypothesis and research question.
        - For numeric types, same plot and table as mentioned above.
        - For categorical types, a bar plot and a table of frequencies for each level and the proportion of that level (percentage).
    - Provide some text for each variable you explore that briefly describes the distribution. The text should be next to the plot and table.
- **Modeling**

  The following sections below apply to regression, classification and time series models. For clustering, you must apply three different types of clustering algorithms, evaluate and explain the goodness of the clustering results, and how the clustering relates to your hypothesis.
- **Model fitting**
    - State at least one hypothesis that can be tested by your models. The hypothesis mentions a relationship you predict between at least one independent variable and the dependent variable.
    - Fit two types of model that include **all of the independent variables*** in the data set as predictors of the dependent variable.
    - For each model:
        - Discuss the relative “importance” of the predictors, where importance is determined by some metric appropriate to that model. For example, linear regression: coefficient and p-value.
        - Mention the overall model fit as appropriate to that model.

        - Briefly describe the fit results in your own words. Include any concerns you have about the fit and how you addressed them. If you have no concerns about the fit results, you don't need to do anything.
        - Explain how the fit results relate to your hypothesis. Typically how the predictor(s) in your hypothesis compared to other predictors as well as the outcome.
    - Briefly compare and comment on the fit results of both models and how they relate to your hypothesis.
- **Model Selection**
    - Based on the fit results above, you may choose to modify the set of predictors you include in your models. If so, explain your rationale for choosing the new predictor set, then fit the two models to the new set of predictors and follow the same steps as listed in the section above.
    - For time series, this would mean choosing the model that best fits the data. Include at least two benchmark models as candidate models.
    - Mention how the results relate to your hypothesis.
- **Model Prediction**
    - **For each model:** Use repeated cross fold validation with 10-folds repeated 3 times with 80% of the data used for training on each fold.
    - State the settings for cv (above) in your own words in the report.
    - Use appropriate metrics, such as RMSE and std dev for regression, or confusion matrix with metrics and an ROC plot with both models for classification
    - Display and explain the prediction results for both models in a table and line plot.
- **Summary and conclusion**
    - Compare and contrast the two models: how they fit the data, their predictions.
    - How did the models support (or not) your hypothesis?
    - Based on the results, what conclusions can you draw about your research question?
    - Mention any threats to validity you can think of.
- **References**
    - List links to your data source(s) and any other sources you used in your project.
    - Use numbers in your report to refer to links in this section, e.g. [1].

* **There are cases where you obtained a dataset with a large number (>10) of predictors, or that includes some predictors that are not relevant to your analysis. In that case, you may exclude predictors from your model. Explain that you did not include some predictors, how many, and why. Note that your models must have at least three predictors. This is not the same as model selection.**

Other Requirements

- Your ***dataset*** cannot be one used in this course, and cannot be synthetic (manually generated). It must have more than 50 observations.
- You submit an Rmd file and the pdf file generated from knitting your Rmd file in a single .zip file. See “Files to submit” below.
- The meta-data section is correctly filled in. Use both first and last names.
- The title is descriptive of your analysis. Do not include any reference to “project 2”, etc.
- Do not include any text that refers to this as a class project.
- The number_sections feature is used to appropriately number headings. Use the pound symbols, #, to appropriately organize the level of your headings.
- All library load statements must be in the top code chunk.
- Use the r.version and seed.val variables in any random process in your Rmd file instead of hard-coded numbers. Note: you need to set the random number generator each time you execute a random process.
- No R code or raw output should appear in the knitted pdf file. Use tables, plots and text to summarize output as needed.
- Do not mention any R-specific functions or features in the pdf file. The report should be completely platform agnostic. The reader should not be able to tell what software was used to create the report.
- Examples of how not to mention R-specific terms:
    NO: “...we used the lm function to…”
    YES: “...we fitted a linear regression model…”
    NO: “...we used the na.omit function to remove any NA values…”
    YES: “...we removed any rows that included missing values…”
- Models: your models must have at least three predictors. Define the dependent and independent variables. Do not display an R formula in the report.
- You must use cross validation in your report
- All plots must be formatted to fit appropriately on the page, captioned and uniquely numbered.
- All tables must be captioned and uniquely numbered.
- No plots or tables are separated by a page break. Use \newpage to avoid this.

Files to Submit

(applies to milestones as well as final submission)

A template Rmd file has been made available for your use. You may use and modify this file as you see fit as long as you fulfill the required sections and other requirements outlined above.

Submit this Rmd file and a pdf file that is the knitted version of the Rmd file. Do not submit any data files. We will only accept an Rmd and a pdf file. Submitting an Rmd file that does not knit will be awarded a maximum of 20% of the project grade.

The files you submit must have this naming convention, replacing the partner1lastname-partner2lastname with your last names:

INFO248-project2-partner1lastname-partner2lastname.pdf
INFO248-project2-partner1lastname-partner2lastname.Rmd

These files must be submitted to the canvas assignment ***in a single archive (zip) file*** named:

INFO248-project2-partner1lastname-partner2lastname.zip

We will not accept any files that do not follow this naming convention. Note: you may use this naming convention for the milestones as well as the final submission.

Plagiarism

We require that your R code and analysis techniques are ***aligned with the examples presented in this class.*** Any variation from these examples ***must be approved by the instructor.*** Submitting code, text, and using other techniques without prior approval will be considered academic dishonesty and may result in severe consequences. Use of AI tools for lookup is OK, but using AI generated output with minimal or no modification is considered plagiarism,an act of academic dishonesty.

Collaboration

You and your partner share the responsibility for the work on the projects in this class. That requires you both to communicate frequently with each other, and to plan the timeline and duties each of you will carry out. If one of you has a particularly challenging week, let your partner know and adjust your work to adapt.

You will individually complete a short assessment of how the collaboration is going before the due date for each project, but if you find that you are having difficulties with your partner, it is important that you contact me directly right away (private piazza post). I will discuss the situation with you in confidence and not take any action without your approval. Please remember that you have a responsibility to your partner to do good work in a timely manner as it will affect both of your grades. ***The instructor reserves the right to modify a project grade based on the level and quality of a student's collaboration.***

Please let the instructor know if you have any questions or concerns about project work at any time.

Grading of Final Project (60pts total):

Note: failure to submit a knitted pdf file or an rmd file that cannot be knitted to pdf will result in a grade of 0. A project submission is required to pass this course (see syllabus).

Formatting	20pts
Analysis	40pts
Total	60pts

Summary of Techniques from the Course:

- Describing how a data is distributed:
    - descriptive stats (numeric data)
    - frequencies (categorical data)
    - plots (also useful presenting modeling results)
- Processing data
    - Missing values
    - Outliers
    - transforming data: log function, scaling, normalizing
    - converting from one type to another
- Comparing distributions (numeric and categorical)
    - hypothesis tests
    - ANOVA
    - contingency tables
    - chi-square
- Models (pick at least two):
    - linear regression (numeric outcome)
    - logistic regression (categorical outcome)
    - KNN (numeric or categorical outcome)
    - decision tree- single, ensemble (numeric or categorical outcome)
    - time series (forecasting the future of a series)
    - clustering (hierarchical, k-means, model-based)
- Modeling techniques
    - goodness of fit, interpreting fit, model selection
    - cross validation
    - evaluating model prediction (numeric, categorical outcomes)